



Sec: SC-60

Coordination compounds

Date: 19-06-2020

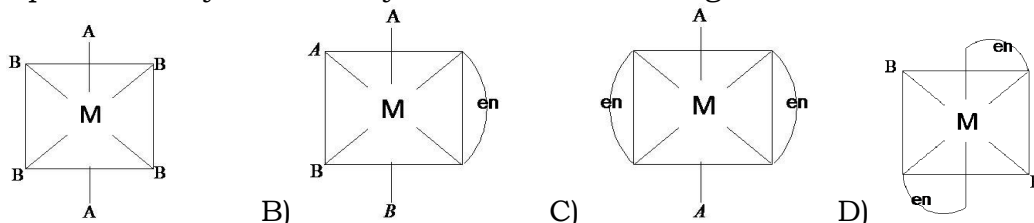
SINGLE ANSWER TYPE

- What is the expected order for increasing octahedral (Δ_0) crystal field splitting for the ligands I^- , F^- , H_2O , NH_3 , en and CO?
 - $I^- < F^- < H_2O < NH_3 < en < CO$
 - $F^- < I^- < NH_3 < en < CO < H_2O$
 - $I^- < F^- < H_2O < CO < NH_3 < en$
 - $CO < en < NH_3 < H_2O < F^- < I^-$
- Tetrahedral complexes of the types of $[Ma_4]$ and $[Ma_3b]$ (where M = metal; a, b = achiral ligands) are not able to show optical isomerism because
 - These molecules/ions possesses C_n axis of symmetry
 - These molecules/ions possesses a plane of symmetry and hence are achiral
 - These molecules possesses a central of symmetry
 - These molecules/ions have non superimposable mirror images
- $[CoCl_2(NH_3)_4]^+ + Cl^- \longrightarrow [CoCl_3(NH_3)_3] + NH_3$
If in this reaction two isomers of the product are obtained, which is true for the initial (reactant) complex:
 - Compound is in cis-forms
 - Compound is in trans-form
 - Compound is in both (cis and trans) forms
 - can't be predicted
- The pair $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$ will be
 - Hydrate isomers
 - Linkage isomers
 - Ionization isomers
 - Coordinate isomers
- The compounds $[Cr(H_2O)_6]Cl_3$, $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ and $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$ exhibit
 - Linkage isomerism
 - Geometrical isomerism
 - Ionization isomerism
 - Hydrate isomerism
- Which ion would you expect to have the largest splitting of d-orbitals?
 - $[Fe(CN)_6]^{4-}$
 - $[Fe(CN)_6]^{3-}$
 - $[Fe(H_2O)_6]^{2+}$
 - $[Fe(H_2O)_6]^{3+}$
- If $\Delta_0 < P$, what is electronic arrangement of metal atom/ion in octahedral complex with d^4 configuration:
 - t_{2g}^4, e_g^0
 - e_g^4, t_{2g}^0
 - t_{2g}^3, e_g^1
 - e_g^2, t_{2g}^2
- Set of d-orbitals which are used by central metal during formation of MnO_4^- ?
 - $d_{x^2-y^2}, d_{z^2}, d_{xy}$
 - d_{xy}, d_{yz}, d_{xz}
 - $d_{x^2-y^2}, d_{xy}, d_{xz}$
 - $d_{x^2-y^2}, d_{z^2}, d_{xz}$
- The total possible coordination isomers for the following compounds respectively are:
 $[Co(en)_3][Cr(C_2O_4)_3]$ $[Cu(NH_3)_4][CuCl_4]$ $[Ni(en)_3][Co(NO_2)_6]$
 - 4, 4, 4
 - 2, 2, 2
 - 2, 2, 4
 - 4, 2, 4
- Select the correct code regarding total number of space isomers for the following compounds:

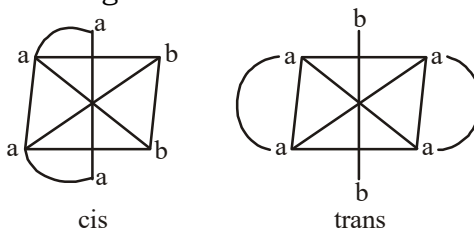
I. $[Ma_3b_2c]^{n+}$	II. $[M(AB)_3]^{n+}$	III. $[Ma_2b_2c_2]^{n+}$
I	II	III
A) 4	4	6
B) 4	3	5

- | | | | |
|----|---|---|---|
| C) | 3 | 3 | 5 |
| D) | 3 | 4 | 6 |

11. $A[M(H_2O)_6]^{2+}$ complex typically absorbs at around 600 nm. It is allowed to react with ammonia to form a new complex $[M(NH_3)_6]^{2+}$ that should have absorption at:
- A) 800 nm B) 580 nm C) 620 nm D) 320 nm
12. The CFSE for $[CoCl_6]^{4-}$ complex is 18000 cm^{-1} . The Δ for $[CoCl_4]^{2-}$ will be:
- A) 18000 cm^{-1} B) 16000 cm^{-1} C) 8000 cm^{-1} D) 2000 cm^{-1}
13. The optical activity is shown by which of the following.



14. Which of the following complex has highest C – O bond energy?
- A) $[Ni(CO)_4]$ B) $[Co(CO)_6]^{3+}$ C) $[Mn(CO)_6]^{+}$ D) None
15. Which complex is likely to show optical activity?
- A) Trans $-[Co(NH_3)_4Cl_2]^+$ B) $[Cr(H_2O)_6]^{3+}$
- C) Cis $-[Co(NH_3)_2(en)_2]^{3+}$ D) Trans $-[Co(NH_3)_2(en)_2]^{3+}$
16. Which of the following statements is correct regarding the chirality (optical isomerism) of the cis and trans isomers of the type $M(aa)_2b_2$ (M stands for a metal, a and b are achiral ligands and aa means bidentate ligands)



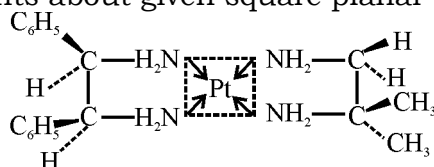
- A) The trans form is achiral and optically inactive while the cis form is chiral and exists in two enantiomeric forms
- B) The cis as well as the trans form are achiral and optically inactive
- C) The trans form is chiral and exists in two enantiomeric forms while the cis form is achiral and optically inactive
- D) The cis as well as the trans form are chiral and each of them exists in two enantiomeric forms
17. How many isomers are possible for the complex ion $[Cr(NH_3)(OH)_2Cl_3]^{2-}$?
- A) 2 B) 3 C) 4 D) 5
18. The correct order for the CFSE (numerical value) for the following complexes is:
- | | | | | |
|--------------------|--------------------|--------------------|---------------------|---------------------|
| Complex | P | Q | R | S |
| Formula | $[CoF_6]^{3-}$ | $[Co(CN)_6]^{3-}$ | $[Co(NH_3)_6]^{3+}$ | $[Co(H_2O)_6]^{3+}$ |
| A) $P > Q > R > S$ | B) $Q > R > S > P$ | C) $S > R > P > Q$ | D) $R > Q > P > S$ | |

19. I : In poly nuclear metal carbonyls, the hybridization of 'c' in terminal carbonyl is 'sp' where as that in bridged carbonyl is 'sp²'
- II : In all mononuclear metal carbonyls, the carbonyls are non-bridged
- III : $Fe(CO)_5$ is diamagnetic and Fe is dsp^3 hybridised
- IV : Metal carbonyls are more stable due to more nucleophilic nature of carbon monoxide.
- The correct statements are:
- A) I, II, III B) I, II, IV C) II, III, IV D) I, II, III and IV

MULTIPLE ANSWER TYPE

20. Which of the following will have three stereo isomeric forms?

- A) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ B) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
 C) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_2\text{Cl}_2]$ D) $[\text{Co}(\text{en})_2\text{ClBr}]$
21. Which of the following complex compounds exhibits cis-trans isomerism?
 A) $[\text{PtCl}_2(\text{NH}_3)_2]$ B) $[\text{PdCl}_2\text{BrI}]$ C) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$ D) $[\text{PtCl}_2\text{BrI}]$
22. Which of the following statements are true for the reaction given below?
 $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$
 A) It is a ligand substitution reaction
 B) NH_3 is a relatively strong field ligand while H_2O is a weak field ligand
 C) During the reaction, there is a change in colour from light blue to dark blue
 D) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, has a tetrahedral structure and is paramagnetic
23. The value of 'spin only' magnetic moment for one of the following configuration in octahedral complex is 2.84 BM. The correct one is/are:
 A) d^4 (in strong field ligand) B) d^2 (in weak field ligand)
 C) d^3 (in weak as well as strong field ligand) D) d^5 (in strong field ligand)
24. Which of the following statements is/are correct?
 A) The stability constant of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is greater than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$
 B) The cyano complexes are far more stable than those formed by halide ions
 C) The stability of halide complexes follows the order $\text{I}^- < \text{Br}^- < \text{Cl}^-$
 D) The stability constant of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is greater than that of $[\text{Cu}(\text{en})_2]^{2+}$
25. Select the incorrect statements about given square planar complex.



- A) It has no geometrical isomer
 B) it is optically active because it does not have plane of symmetry
 C) It is optically inactive because square planar complex have plane of symmetry
 D) It is optically active because it has symmetric center
26. A complex of molecular formula $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ can show geometrical isomerism. It can be correctly represented as
 A) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ B) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
 C) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ D) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
27. Which of the following combinations is/are correct?
 A) $\text{K}[\text{Ag}(\text{CN})_2]$ - show linkage isomerism
 B) $[\text{Ag}(\text{NH}_3)_2][\text{CuCl}_4]$ - show coordination isomerism
 C) $[\text{CrCl}(\text{NH}_3)_5]\text{SO}_4$ - show ionization isomerism
 D) $[\text{PtCl}_4\text{I}_2]$ - show optical isomerism
28. Select correct statement(s):
 A) $\text{Mn}_3(\text{CO})_{12}$ is Dodecacarbonyltrimanganese(0)
 B) EDTA is a polydentate, flexidentate and chelating ligand
 C) $[\text{Co}(\text{H}_2\text{O})_6]^{+2}$ is more stable than $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$
 D) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ have dsp^2 hybridisation
29. In which pairs, both the compounds show geometrical isomerism?
 A) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Co}(\text{en})_2\text{Cl}_2]$ B) $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^{2-}$ and $[\text{CoCl}_2(\text{OX})_2]^{3-}$
 C) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Cr}(\text{en})_3]^{3+}$ D) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ and $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
30. Select the correct statements:
 A) Chelation effect is more important for penta and hexadentate ligands for stability of complex
 B) Greater the charge on the central metal cation, greater the value of Δ (CFSE)
 C) In complex ion $[\text{CoF}_6]^{3-}$, F^- is a weak field ligand, so that $\Delta_{\text{oct}} < P$ (Pairing energy) and it is low spin complex

- D) $[\text{CoCl}_2(\text{NH}_3)_2(\text{en})]^+$ complex ion will have four different stereoisomers
31. Which statement(s) is/are correct?
- A) $[\text{Ag}(\text{NH}_3)_2]^+$ is linear with sp hybridization of Ag^+ ion
- B) NiCl_4^{2-} , VO_4^{3-} , MnO_4^- have tetrahedral geometry
- C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{CN})_4]^{2-}$ have dsp^2 hybridization of the metal ion
- D) $\text{Fe}(\text{CO})_5$ have bipyramidal structure with dsp^3 hybridization of iron
32. Identify the correct statements:
- A) Δ_0 increasing order is $[\text{CrCl}_6]^{3-} < [\text{Cr}(\text{NH}_3)_6]^{3+} < [\text{Cr}(\text{CN})_6]^{3-}$
- B) CFSE for $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is $-0.4\Delta_0$
- C) $[\text{NiCl}_4]^{2-}$ as well as $[\text{Ni}(\text{CO})_4]$ are paramagnetic
- D) The halide ions are arranged as $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$ in the spectro chemical series.
33. Carbon monoxide is a very common ligand in organometallic chemistry. It is particularly good at stabilizing low oxidation states of central metal. The correct statements about metal carbonyls are:
- A) C - O bond length in the increasing order is
 $[\text{Mn}(\text{CO})_6]^+ < [\text{Cr}(\text{CO})_6] < [\text{V}(\text{CO})_6]^- < [\text{Ti}(\text{CO})_6]^{2-}$
- B) M - C bond length is of the order $[\text{Cr}(\text{CO})_6] > [\text{Cr}(\text{CO})_5(\text{PPh}_3)] > [\text{Cr}(\text{CO})_5(\text{PEt}_3)]$
- C) Among $[\text{Mn}(\text{CO})_4]^{-3}$, $[\text{Pb}(\text{CO})_6]^{2-}$, $[\text{Cr}(\text{CO})_6]$ and $[\text{Ni}(\text{CO})_4]$, the most unstable carbonyl is $[\text{Pb}(\text{CO})_6]^{2-}$
- D) CO is a π - acid ligand

INTEGER TYPE

34. Find out the numbers of isomers (structural and stereo) produced by the complex $[\text{Pt}(\text{NH}_3)_2(\text{NO}_2)\text{Cl}_3]$
35. How many of the following complexes can exist as 'd' and 'l' isomers
 $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$, $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, trans $[\text{Co}(\text{en})_2\text{Cl}_2]^+$,
 Cis $[\text{Co}(\text{en})_2\text{Cl}_2]^+$, $[\text{Cr}(\text{Ox})_3]^{3-}$, $[\text{Cr}(\text{en})_3]^{3+}$, $[\text{Pt}(\text{en})_2\text{Cl}_2]$

MATRIX MATCHING TYPE

- | | |
|--|--|
| 36. Set - I (Comparison of stability) | Set - II (Responsible factors) |
| 1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} > [\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ | (P) Principal quantum number of d subshell |
| 2) $\text{Co}[\text{Hg}(\text{SCN})_4] > \text{Hg}[\text{Co}(\text{SCN})_4]$ | (Q) Charge density of central metal atom |
| 3) $[\text{Cr}(\text{CN})_6]^{3-} > [\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ | (R) Strength of ligand |
| 4) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-} > [\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ | (S) Chelation |
| A) 1 - Q, 2 - R, 3 - P, 4 - S | B) 1 - P, 2 - Q, 3 - S, 4 - R |
| C) 1 - R, 2 - Q, 3 - S, 4 - P | D) 1 - Q, 2 - P, 3 - R, 4 - S |
| 37. Set - I (Complex Compound) | Set - II (Type of Isomerism Shown) |
| 1) $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$ | (P) Geometrical isomerism |
| 2) $[\text{Co}(\text{en})_3]\text{Cl}_3$ | (Q) Optical isomerism |
| 3) $[\text{Co}(\text{en})_2(\text{NO}_2)_2]\text{Cl}$ | (R) Linkage isomerism |

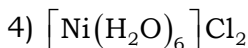
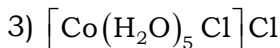
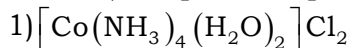


(S) Ionisation isomerism

A) 1 – P, 2 – Q, 3 – R, 4 – S

C) 1 – P, 2 – Q, 3 – S, 4 – R

38. Set – I (Complex compound)



state

B) 1 – Q, 2 – P, 3 – S, 4 – R

D) 1 – R, 2 – Q, 3 – S, 4 – P

Set – II (Related property)

(P) Give precipitate by the action of $AgNO_3(aq)$

(Q) Paramagnetic in nature

(R) Exhibit geometrical isomerism

(S) Does not conduct electricity in its aqueous

A) 1 – P, 2 – S, 3 – R, 4 – Q

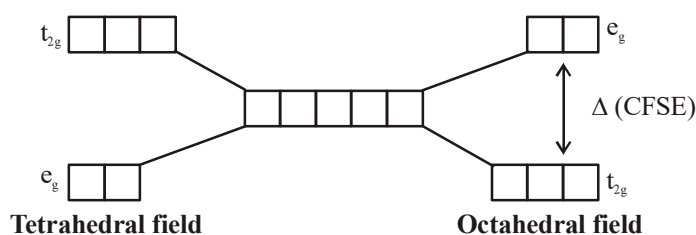
C) 1 – S, 2 – R, 3 – Q, 4 – P

B) 1 – R, 2 – S, 3 – P, 4 – Q

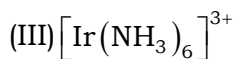
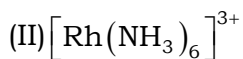
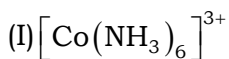
D) 1 – R, 2 – P, 3 – S, 4 – Q

PASSAGE-1

When the degenerate orbitals of d sub-shell of an isolated atom / ion come under influence of magnetic field of ligands, the degeneracy is lost. The two sets t_{2g} (d_{xy} , d_{yz} , d_{zx}) and e_g (d_{z^2} , $d_{x^2-y^2}$) are either higher or lower energetic depending upon the nature of magnetic field.



Value of CFSE depends upon nature of ligand and spectro chemical series has been made experimentally. For tetrahedral complexes, Δ_t is about 4/9 times to Δ_o . This energy lies in visible region and that is why electronic transitions are responsible for colour. Such transitions are not possible with d^0 and d^{10} configuration.

15. The values of CFSE (Δ_o) for complexes given below follow the order:

A) I < II < III

B) I > II > III

C) I < II > III

D) I = II = III

16. The d – orbitals which are lower energetic in an octahedral magnetic field are:

A) d_{xy} and d_{z^2} B) $d_{x^2-y^2}$ and d_{z^2} C) d_{xy} , d_{xz} and d_{yz} D) d_{z^2} only**PASSAGE-2**

In metal carbonyls, there is synergic bonding interaction between metal and carbon monoxide. This leads to increase in strength of metal-ligand bond and decrease in bond order of CO in carbonyl complex as compared to bond order in carbon monoxide. Simple carbonyls are invariable spin-paired complexes except for vanadium metal.

17. Which of the following statement is false for Nickel carbonyl $[Ni(CO)_4]$?

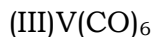
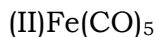
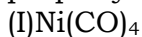
A) It is a colourless compound

B) The Ni – C – O group is linear

C) The four carbonyl groups are lying at the corners of a regular tetrahedron

D) The metal – carbon bond length (for σ bond) does not alter

18. Which amongst the following metal carbonyls are inner orbital complexes with diamagnetic property?



A) I and II only
B) II, III and IV only
C) II and IV only
D) I, II and IV only

B) II, III and IV only
D) I, II and IV only

KEY

[illegible]